Clinical question 10: What are the recommendations for laboratory monitoring during treatment with isotretinoin?

The **European Medicines Agency (EMA)** recommends laboratory testing of **lipids and liver enzymes** before initiating isotretinoin treatment, and at **1 month and 3 months** after starting therapy. More frequent monitoring may be warranted in patients with underlying comorbidities.

The **European guidelines** align with the EMA, advising assessment of liver enzymes and lipid levels **before treatment**, at **1 month**, and then **every 3 months** thereafter. However, published guidelines generally lack detailed recommendations on laboratory monitoring beyond pregnancy testing.

The decision to perform laboratory monitoring should be individualized, based on the **treating physician's clinical judgement**, taking into account the patient's **risk factors, medical history**, and any **emerging symptoms**.

Further evidence

A **systematic review and meta-analysis (2016)** evaluated studies on laboratory monitoring during isotretinoin treatment for acne, focusing on lipids, liver function tests, and complete blood counts. While isotretinoin was associated with **statistically significant changes** in mean values of white blood cells, liver enzymes, and lipid panels, these changes were not considered clinically high-risk. Only a **small proportion** of patients developed abnormal results. The authors concluded that **routine monthly laboratory testing is not justified** for all acne patients receiving standard-dose isotretinoin.

A **retrospective study of 515 patients** (574 treatment cycles) found: - **Leucopenia** in 2.4% of cycles, none requiring treatment discontinuation - **Non-clinically significant thrombocytopenia** in 1.6% of cycles - **Elevated alanine aminotransferase (ALT)** in 3.3% of cycles, without need for discontinuation - **Lipid abnormalities** were more common - **67.7% of all laboratory abnormalities** occurred within the first **60 days** of treatment

The study proposed a **risk-adapted monitoring strategy**: - Assess **personal and family history** of lipid, liver, or hematologic disorders - For **healthy patients**: - Perform baseline testing of **lipids and liver function** - Repeat testing at **month 2**, after reaching maximum daily dose - If results are normal, **further testing may not be necessary** - If **abnormalities are present** or there is a **known lipid disorder**, continued monitoring is advised - **Complete blood count (CBC)**: routine testing not recommended unless there is a known hematologic condition or risk factor

This study has been considered **practice-changing**, highlighting the issue of **overmonitoring** in isotretinoin-treated acne patients. Adopting this approach could reduce **unnecessary blood draws** and **healthcare costs**.

Comments and conclusions

Clinical guidelines aim to optimize patient care based on the best available evidence regarding treatment **efficacy and safety**. However, limitations exist in both the evidence base and guideline quality.

As noted in the **US work group guidelines**, adherence does not guarantee successful outcomes. Guidelines should not be interpreted as establishing a **standard of care**, nor should they exclude other reasonable approaches. The final decision on therapy must be made collaboratively by the **physician and patient**, considering individual circumstances and disease variability.

The **European guidelines** emphasize that clinicians may be held **medically liable** if they deviate from recommended prescribing practices in the event of adverse outcomes.

A separate evaluation of acne guidelines published between **2013 and December 2016**, using the **AGREE II** tool and **Institute of Medicine** criteria for trustworthiness, found significant deficiencies in guideline quality. Improvements are needed to meet modern standards for guideline development.

Notably, there is **considerable variation** among current guidelines. A key step toward harmonization would be the adoption of a **common classification system for acne**.